A Tiny Motor Inspired by an Ancient Screw

From: ScienceDaily Top Technology Mined: 2026-11-02 3 min read

Scientists have built a tiny motor that spins when electricity runs through it. If they switch the **voltage**, it spins the other way.

The motor copies an old machine called the **Archimedes screw**. Long ago, people used it to lift water uphill. It uses a **spiral** that turns. In the new motor, tiny bits called **electrons** flow through a **spiral molecule**. As they flow, they make it turn.

The **spiral molecule** was made by Prof. Michael Pittelkow's team at the University of Copenhagen. It is only a few millionths of a millimeter big. At the Weizmann Institute in Israel, Prof. Oren Tal and Dr. Anil Kumar Singh put **molecules** between two gold tips. They could stretch or squeeze the tiny screw.

When **voltage** was added, the **molecule** turned. Switching the **voltage** turned it the other way. It is the first motor run by current that is firmly joined to two metal ends and can turn in a direction scientists choose.

Turning brought a surprise. The spinning **molecule** carried electricity up to 100 times better than when it was still.

Word Treasure

voltage -- The force that makes electricity flow through a wire.
electrons -- Tiny bits that carry electricity.
spiral -- A shape that curls around like a spring.
molecule -- A tiny piece of matter, far too small to see.
Archimedes screw -- An old machine with a turning spiral that lifts water uphill.

Background you need

Machines can be far too small to see, built from single molecules.

Electricity can make tiny parts move.

Scientists copy old machines, like the Archimedes screw, in new tiny forms.

Break it down

WHO: Scientists, including Prof. Michael Pittelkow's team and Prof. Oren Tal and Dr. Anil Kumar Singh

WHAT: They built a tiny motor that spins on electricity

WHERE: At the University of Copenhagen and the Weizmann Institute in Israel

WHY: They wanted a motor that spins when electricity runs through it and reverses on command

Why it matters

Tiny machines like this could one day power tools far smaller than anything you can see.

Test what you remember

1. What makes the motor spin?

- (A) Gold tips
- (B) Sulfur atoms
- (C) Wind
- (D) Electricity

2. What happens when the voltage is switched?

- (A) It stops
- (B) It carries less electricity
- (C) It spins the other way
- (D) It grows bigger

3. What old machine does the motor copy?

- (A) A gold tip
- (B) A sulfur atom
- (C) Archimedes screw
- (D) A microscope

4. How big is the spiral molecule?

- (A) A millimeter
- (B) 100 times bigger
- (C) A few millionths of a millimeter
- (D) Two gold tips wide

5. What surprise did turning bring?

- (A) It carried electricity up to 100 times better
- (B) It broke apart
- (C) It stopped spinning
- (D) It got colder

6. Who made the spiral molecule?

- (A) Prof. Michael Pittelkow's team
- (B) Dr. Anil Kumar Singh
- (C) The Weizmann Institute
- (D) Prof. Oren Tal

Answer key (try the quiz first!): 1: D 2: C 3: C 4: C 5: A 6: A

Questions to think about

MY THOUGHTS (at least 20 words):